

CYBER-1 INTRUSION DETECTION

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Project Goal:

Our project goal is to use artificial intelligence (AI) to enhance the capability of the IDS to monitor and alert system administrators of any anomalies in their network traffic. We chose this business model due to the ever-increasing IoTs (internet of things) and IoE (internet of everything) devices as well as the exponential growth in network traffic being targets of malevolent actors who want to exploit any vulnerability possible for many reasons like financial gain, revenge, stealing of intellectual property, etc. This project to use UNSW-NB 15 dataset to train algorithms to learn from the network data which involves training AI models to identifies the pattern of normal behavior and spot unusual or suspicious activities within the Network intrusion dataset that is UNSW-NB15 dataset. This goal includes the training and testing different AI/ML classifiers from the dataset, testing their accuracy, and evaluating model's performance based on precision and recall methods, Saving and loading the best trained model for easy reuse without repeating the whole training process.

Exploratory Data Analysis (EDA):

The EDA is a strategy to analyze and the investigation of the dataset to understand its insight details before splitting it for testing and training the dataset. The EDA is important so that the dataset can be analyzed to look for any patterns or relationships in the data before applying the algorithm. The EDA is basically visualizing the data to get an idea of how it is trending.

Analyze dataset features:

We are using the UNSW-NB15 data set to train our AI model. The UNSW-NB 15 data set has been a popular dataset for enhancing IDS capabilities in detecting network traffic anomalies at a higher rate and lowering the rate of false alarms (false positive) rates (Tahri, 2022). We are using this dataset because it is a comprehensive dataset that provides results in the form of normal or abnormal behaviors. Abnormal behaviors in traffic will be encoded with 1 and normal behavior will be encoded with 0.

Statistical features		16 hours	15 hours
No. of flows		987,627	976,882
Src_bytes		4,860,168,866	5,940,523,728
Des_bytes		44,743,560,943	44,309,195,509
Src_Pkts		41,168,425	41,829,810
Dst_pkts		53,402,915	52,585,462
Protocol types	TCP	771,488	720,665
	UDP	301,528	688,616
	ICMP	150	374
	Others	150	374
Label	Normal	1,064,987	1,153,774
	Attack	22,215	299,068
Unique	Src_ip	40	41
	Dst_ip	44	45

Table 1 Dataset features (Moustafa & Slay, 2015)

Refer above Table 1 for detail feature list. The UNSW-NB15 dataset has 9 categories of attack that it focuses on: Normal (indicated by 0), Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shell Code and Worms. The dataset is robust in that it uses three networks and 45 specific IP addresses (Moustafa & Slay, 2015). It is often difficult to choose a classification that can accurately and effectively improve the performance of an IDS so different algorithms for the various types of attacks will need to be evaluated (Tahri, 2022). There are various classification algorithms that can be used with the UNSW-NB15 dataset including Decision Trees, Logistic Regression, Support Vector Machine, Random Forest, Naïve Bayes, and K-Nearest Neighbor.

EDA methods:

We use different EDA methods to analyze and understand the characteristics of each variable, various patterns and to know the insights of the datasets. The EDA methods use in the project are as below:

1. Label Encoder:

The label encoder is used to convert the text data to number format which makes it easier for computer systems to understand and work better with the classification of data in machine learning. The label encoder usually assigns random number to the text but to get more detailed analysis we used custom label encoder which assign number from 1 and go in chronological order which helps to read and understand data from the graphs and figures for further analysis. It helps in numerical feature analysis. In the UNSW_NB15 dataset, we used custom label encoder for three columns, proto, services and state as it has text data. The converted columns are named “proto_encoded”, “service_encoded” and “state_encoded”.

2. Histogram

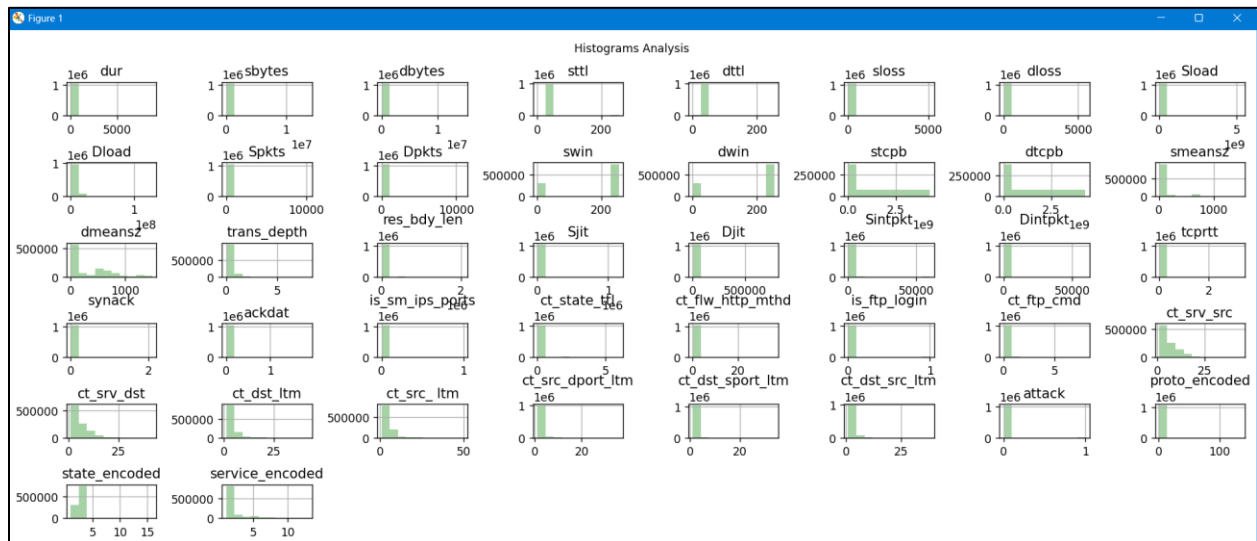
The histogram shows the data values grouped together and put into different classes. It shows how frequently data in each class shows up in the dataset. It shows the frequency of different points of data, the location of that data and the overall spread of the dataset. It shows the variance and any data that is skewed. It can also detect outliers in the dataset. The distribution of the data can be determined using the histogram. In the dataset, there are total 42 different columns and those are as below:

Acronym	Full forms	Acronym	Full forms	Acronym	Full forms
proto	Protocol	sloss	Source Loss	swin	Source Window
state	State	dloss	Destination Loss	dwin	Destination Window
dur	Duration	service	Service	stcpb	Source TCP Base Sequence Number
sbytes	Source Bytes	Sload	Source Load	dtcpb	Destination TCP Base Sequence Number
dbytes	Destination Bytes	Dload	Destination Load	smeansz	Source Mean Packet Size
sttl	Source Time To Live	Spkts	Source Packets	dmeansz	Destination Mean Packet Size
dttl	Destination Time To Live	Dpkts	Destination Packets	trans_dept h	Transaction Depth
res_bdy_len	Response Body Length	ackdat	ACK Data Time	ct_srv_dst	Connection Server Destination Count
Sjit	Source Inter-Packet Jitter	is_sm_ips_ports	Is Source Port Equal To Destination Port	ct_dst_ltm	Connection Destination Count Last Time

Djit	Destination Inter-Packet Jitter	ct_state_ttl	Connection State TTL	ct_src_ltm	Connection Source Count Last Time
Sintpkt	Source Interarrival Time	ct_flw_http_mthd	Connection Flow HTTP Methods	ct_src_dport_ltm	Connection Source Destination Port Count Last Time
Dintpkt	Destination Interarrival Time	is_ftp_login	Is FTP Login?	ct_dst_sport_ltm	Connection Destination Source Port Count Last Time
tcprrt	TCP Round-Trip Time	ct_ftp_cmd	Connection FTP Commands	ct_dst_src_ltm	Connection Destination Source Count Last Time
synack	SYN-ACK Time	ct_srv_src	Connection Server Source Count	attack	Indicates an Attack or Not

(Table 2 Variables in the UNSW-NB15 dataset)

The columns listed in (Table 2, show its representation of each class in the dataset as shown in (Figure 1



(Figure 1, Histogram Analysis)

As the UNSW-NB 15 is very large dataset and training and testing the models using this large dataset is very time consuming and to avoid that based on the representation of each variable, its pattern and behavior shown in (Figure 1, we reduced the number of the columns to 16 and those are:

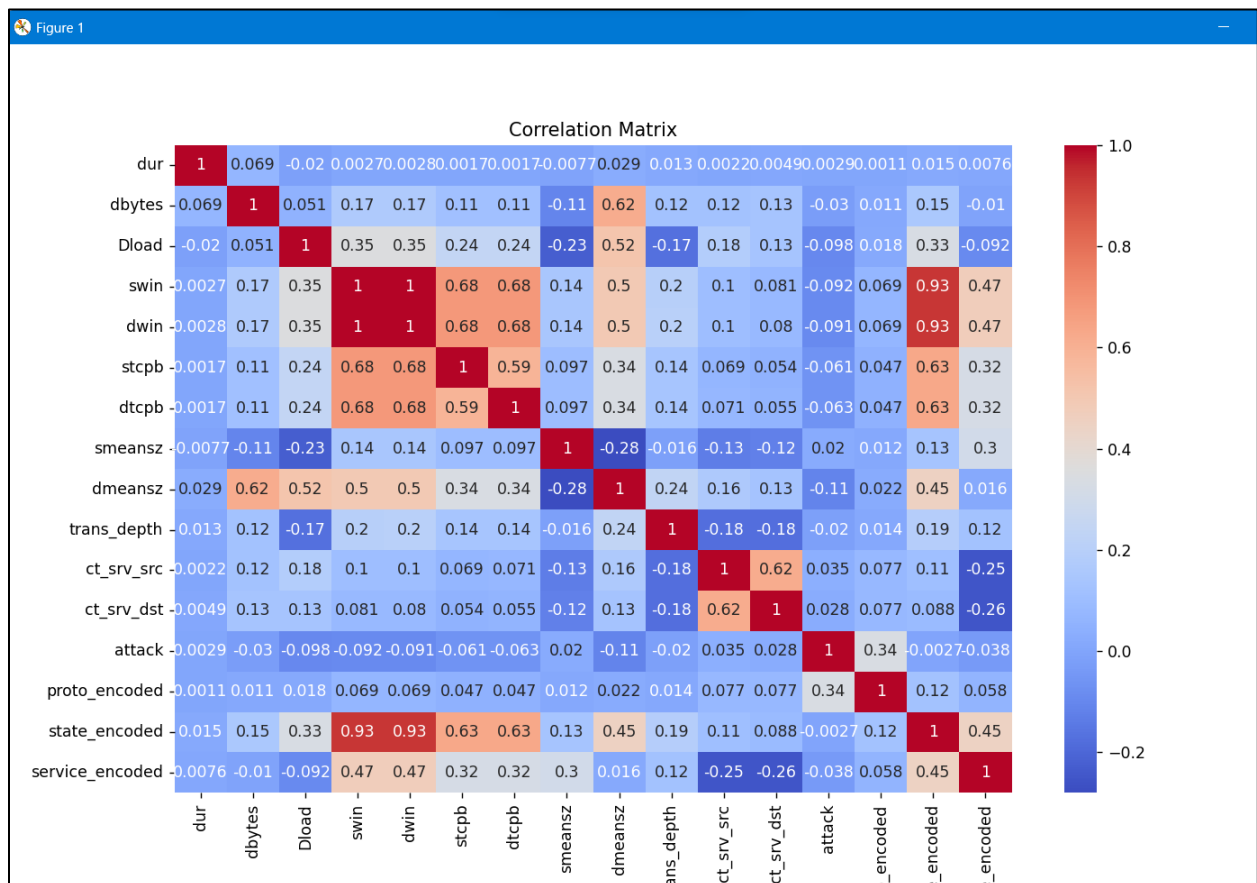
dur	Duration	dbytes	Destination Bytes
dmeansz	Destination Mean Size	attack	Indicates if the traffic is an Attack
Dload	Download Size (or Data Load)	stcpb	Source TCP Base (or Source TCP Bytes)
trans_depth	Transaction Depth	proto_encoded	Protocol Encoded
swin	Source Window (TCP Window Size in Source)	dtcpb	Destination TCP Base (or Destination TCP Bytes)

ct_srv_src	Count of Server Source	state_encoded	State Encoded (TCP Connection State)
dwin	Destination Window (TCP Window Size in Destination)	smeansz	Source Mean Size
ct_srv_dst	Count of Server Destination	service_encoded	Service Encoded

(Table 3 selected columns from the dataset)

3. Correlation Matrix:

The diagonal elements are represented by the total correct values predicted for each class. The lighter the color the greater the number. Lighter colors on the diagonal elements and darker on the rest of them mean our model is either performing well or not based on the colors.

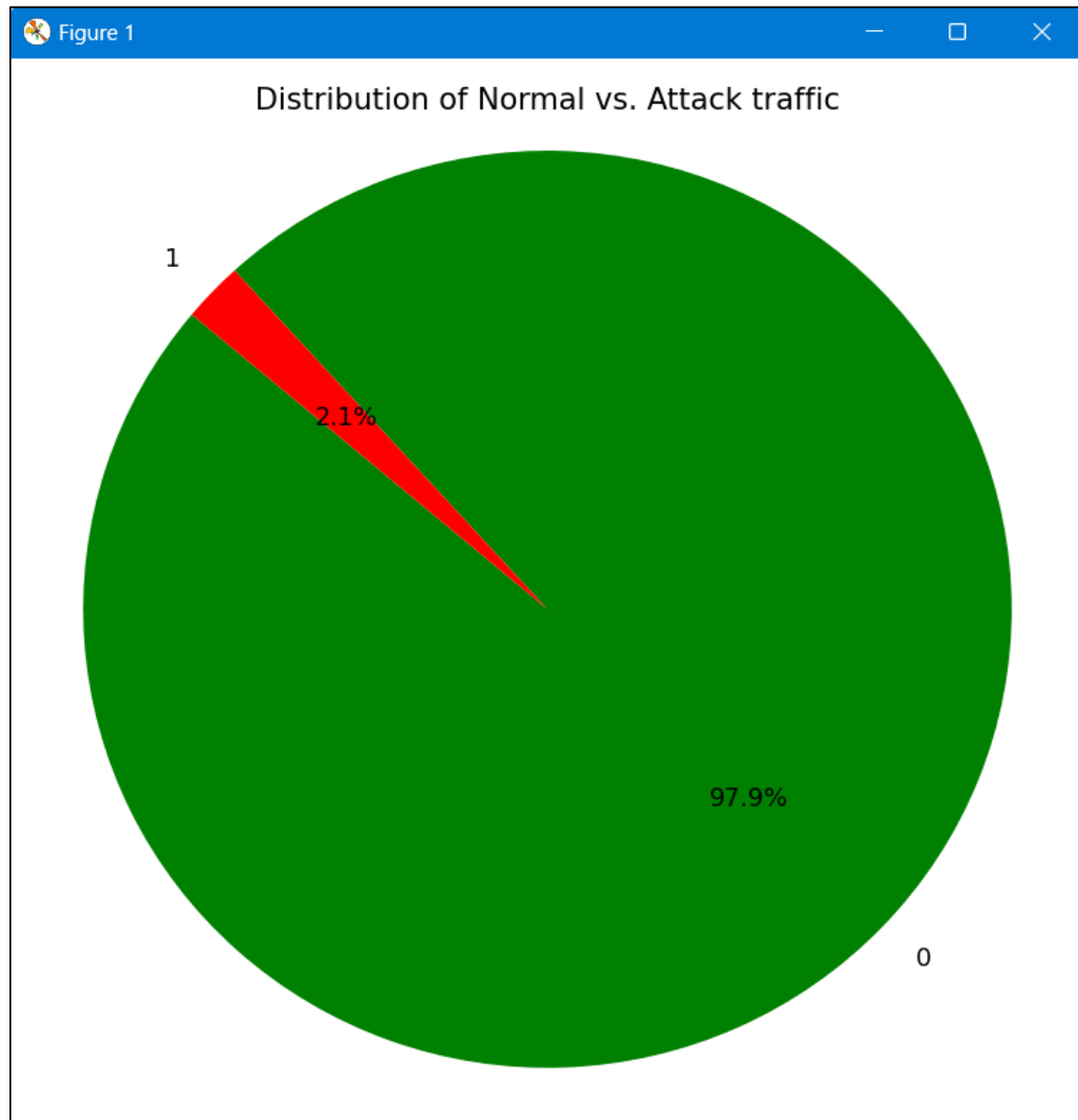


(Figure 2, Correlation matrix)

In Correlation matrix, it shows the relationship between the variables in the dataset which shows the how each variable related to each other by using the values between -1 to 1. Here value closer to 1 represents a strong positive relationship, for example in (Figure 2, "swin" and "state_encoded" shows positive relationship. The value closer to -1 shows a strong negative relationship, for example, "dbytes" and "service encoded" shows a strong negative relationship and around 0 implies little to no relationship between variables.

4. Pie Chart

The pie chart shows the breakdown of the data categories. Each category represented shows the percentage of data points that category has. The pie chart helps organize and how the data. One disadvantage of the pie chart can be that it does not reveal exact values.



(Figure 3 Pie chart)

The above pie chart shows, 2.1% of attack data is distributed all over the dataset and analyzing this amount of data in this large dataset is a complicated task.

5. Numerical Feature Analysis

The numerical feature analysis shows the relevant features for the classification model. It cuts down noise in the data by including and excluding important features without altering them. This can reduce the size of the input data without affecting the performance. Below are the 16 graphs of the numerical feature data.

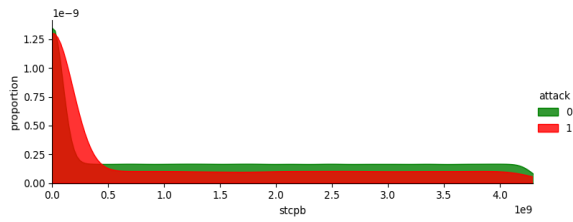


Figure 4, Numerical analysis of stcpb

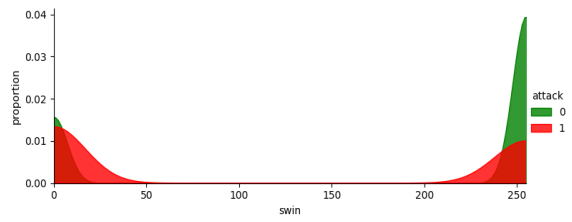


Figure 5, Numerical analysis of swin

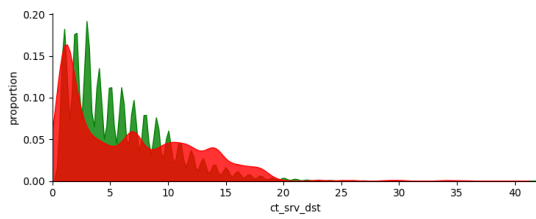


Figure 6, Numerical analysis of ct_srv_dst

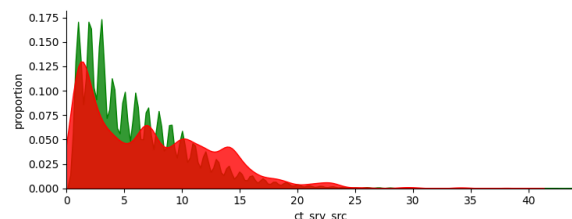


Figure 7, Numerical analysis of ct_srv_src

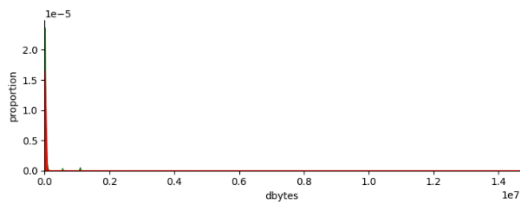


Figure 8, Numerical analysis of dbytes

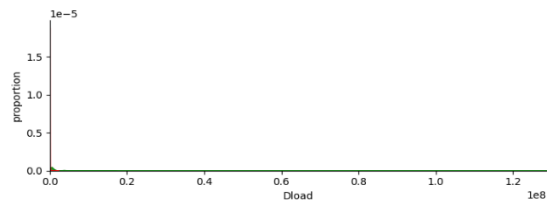


Figure 9, Numerical analysis of dload

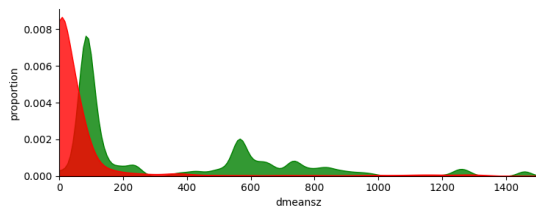


Figure 10, Numerical analysis of dmeansz

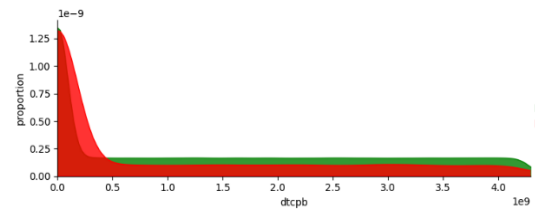


Figure 11, Numerical analysis of dtcpb

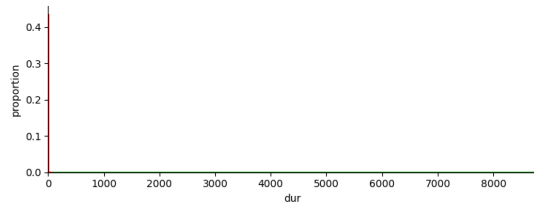


Figure 12, Numerical analysis of dur

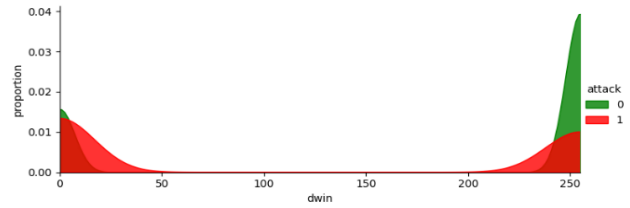


Figure 13, Numerical analysis of dwin

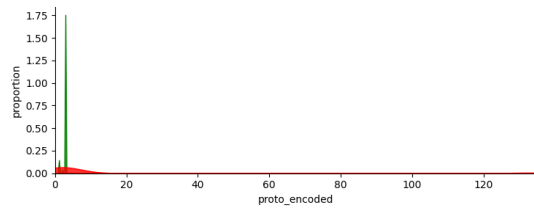


Figure 14, Numerical analysis of proto_encoded

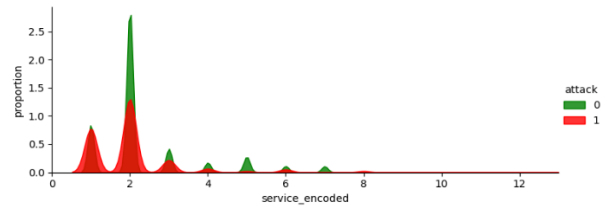


Figure 15, Numerical analysis of service_encoded

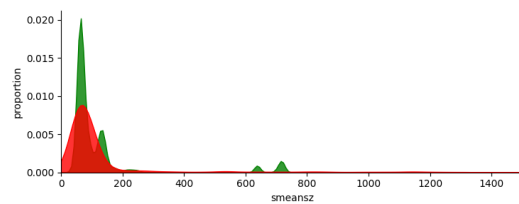


Figure 16, Numerical analysis of smeansz

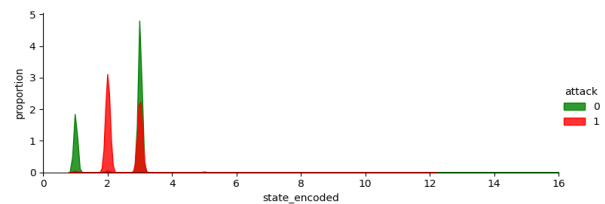


Figure 17, Numerical analysis of state_encoded

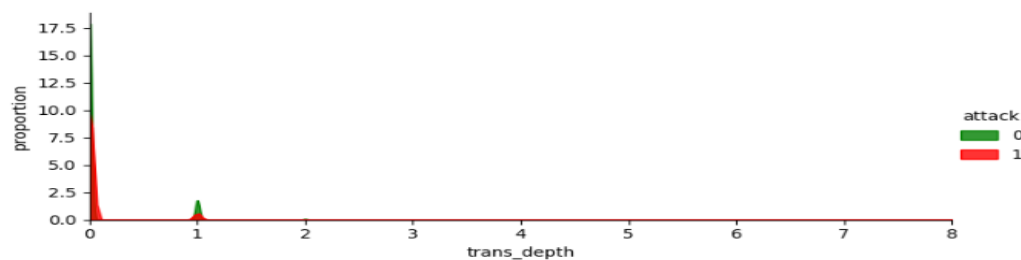


Figure 18, Numerical analysis of trans_depth

6. Scatter Matrix

The scatter matrix is a pair-wise plot of several variables in a matrix format. This is used to determine whether correlation exists between the variables and whether that correlation is positive or negative. This shows the distribution of the single variables and the relationships between variables in the dataset. A positive correlation means that the two variables are moving in the same direction (if one increases, then the other will increase). Negative correlation means that the two variables are moving in the opposite

direction. Zero correlation would mean that the data points will have no relation and no best fit line will be ascertained.

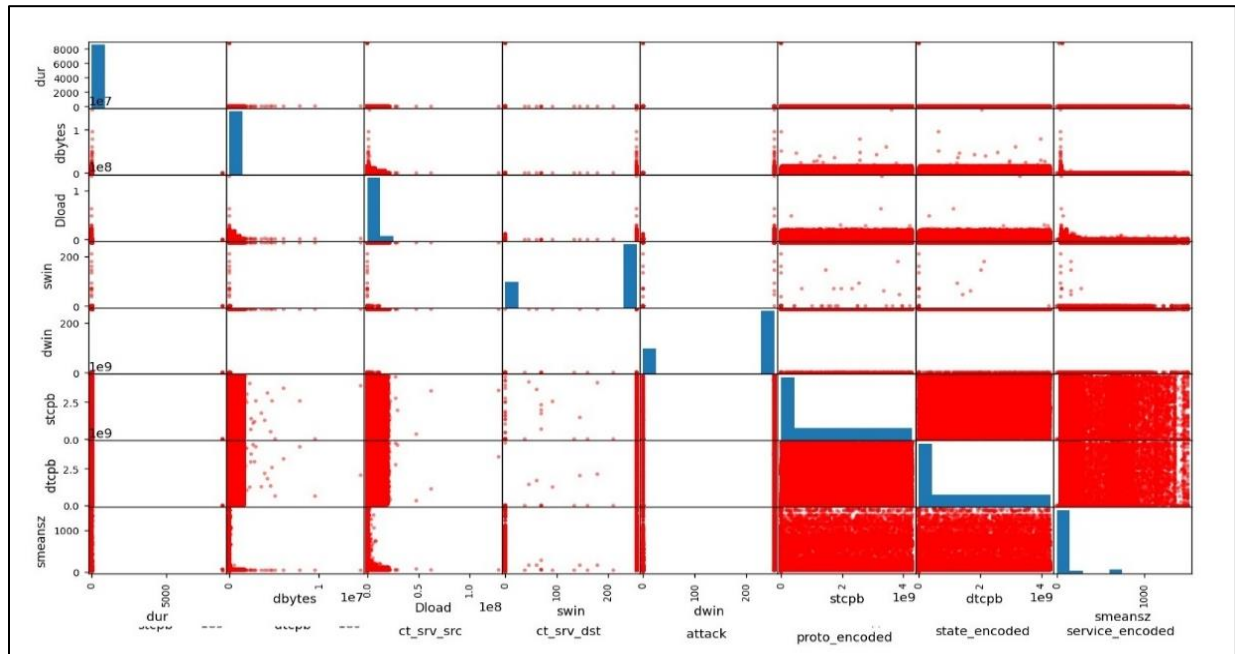


Figure 19, Scatter matrix of dataset (1)

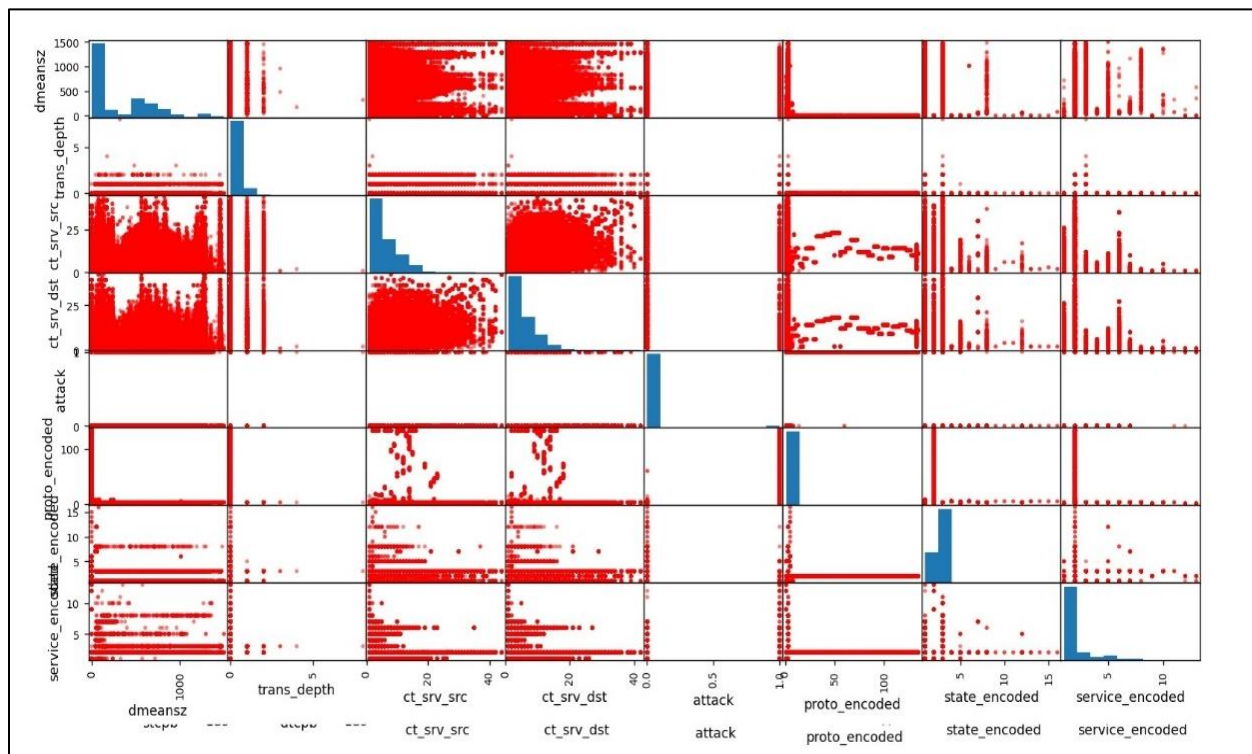


Figure 20, Scatter matrix for dataset (2)

Why you chose these methods [2]

Random Forest:

Random forest algorithm is commonly used and combines the output of multiple decision trees to formulate a single result. A decision tree is usually in the form of a question and can have a series of questions (IBM). In our case, the decision tree consists of, “is this a cyber-attack?” or “is this not a cyber-attack?” The term “forest” in its name is generated and trained through “bagging” or “bootstrap aggregating” which improves the accuracy of the machine learning algorithm.

Selection Choice: We chose the random forest algorithm due to its flexibility, precision and ease when working with large amounts of data.

Logistic regression:

Logistic regression is used for classifying tasks that have a probability in a statistical format and analyzes the relationships between independent variables and dependent binary variables. In logistic regression the outcome must be a yes or no, 0 or 1, or true or false and the probabilistic value will be between 0 and 1.

Selection Choice: We chose to test with the logistic regression algorithm due to its simplicity and ease of implementation. It allows for data to be updated more easily and the outputs are more calibrated than other models (community, 2023).

While both algorithms have their benefits, we chose to show these two as they are different from each other in many ways and allow us to see different results. For example, random forest provides classification and regression while logistic regression is only for binary classification. Random forest does not require feature scaling while logistic regression does, and random forest is more accurate whereas logistic regression is less accurate but works faster (community, 2023).

k-Nearest Neighbor (kNN):

KNN is a machine learning algorithm that can be used in supervised as well as unsupervised machine learning. We used it on our dataset to train a supervised model. This algorithm is based on the standard Euclidean metric, which is used to calculate distance, (Kasongo, 2020)

Selection Choice: KNN is known for its high precision accuracy, ease of implementation and it can be utilized for both small and large data however it can be slow to run and can cause outliers for large amounts of data. We chose to use it to get a bigger picture and get an overall better picture of which algorithm works for this dataset.

Results discussion [2]

We used the cross validation, confusion matrix, normalized confusion matrix, the RoC curve and precision recall curve to evaluate our model:

The best classification algorithm will be the one that has the best performance based on the confusion matrix and accuracy. The confusion matrix evaluates the instances where the model either correctly or incorrectly classified the data. There are four results that the confusion matrix will give: True positive (correct detection), false positive (normal is incorrectly classified as attack data), true negative (normal traffic correctly identified) and false negative (intrusion was not identified). We intend to test the above models to determine which model is the most accurate and based on the metrics from the confusion matrix before deployment of the AI model to ensure validity of the model.

		Actual values	
		Positive (1)	Negative (0)
Predicted values	Positive (1)	True Positive	False Positive
	Negative (0)	False Negative	True Negative

Figure 21, Confusion Matrix (Narkhede, 2018)

Predictor Accuracy

Each model must be checked for accuracy and the predictor accuracy seeks to give output that shows how well the model is performing when making predictions. A model can be safely deployed if it can achieve a high threshold. So, basically it is a representation of the fraction of correct predictions. Figures 6, 7 and 8 reflect predictor accuracies for KNN, Random Forest, and Logistic regression.

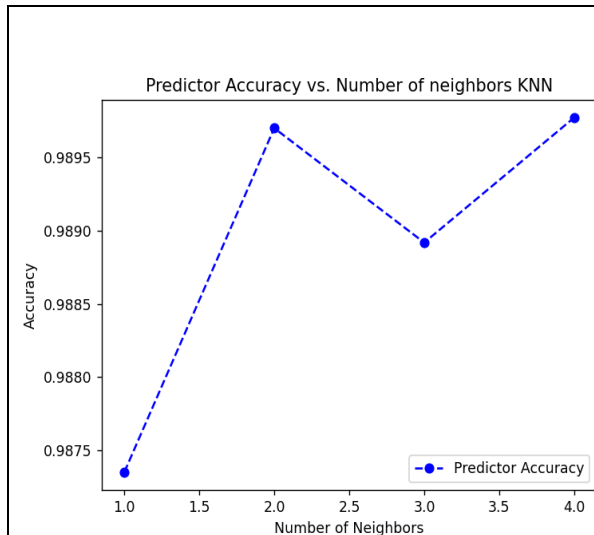


Figure 22, KNN

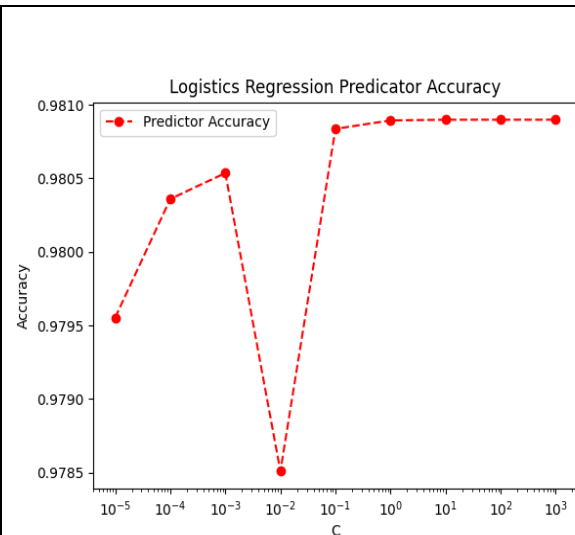
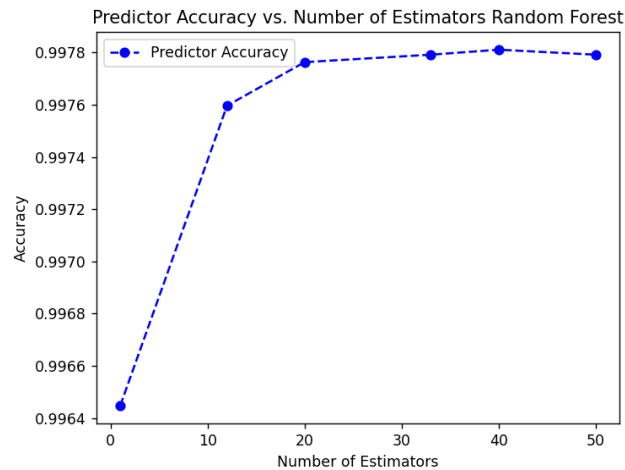
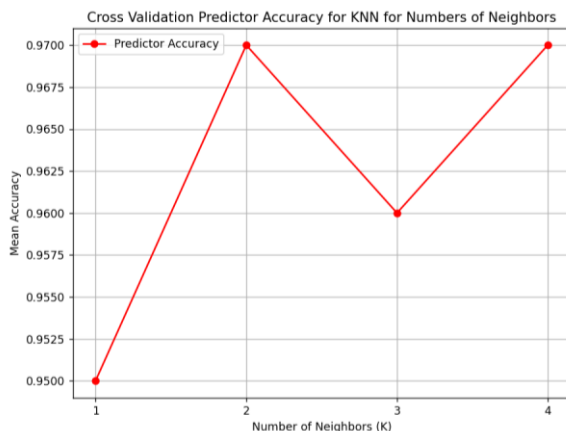
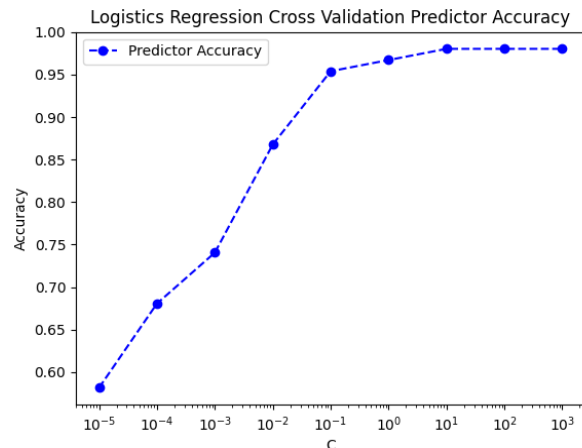


Figure 23, Logistics Regression

*Figure 24, Random Forest*

Cross Validation Predicted Accuracy

Cross validation predicted accuracy is used to test the performance of the model. This is done by using a separate amount of data from the dataset that the model has not been trained on and then comparing how well the model did at predicting correctly. Our model was used for training, and one used for testing. Cross validation also reduces the chance of overfitting. Overfitting is machine learning behavior that happens when the model will make accurate predictions on training data but will not make accurate predictions when introduced to new data. The following figures show the cross-validation results of KNN, Random Forest, and Logistic Regression.

*Figure 25, KNN**Figure 26, Logistics Regression*

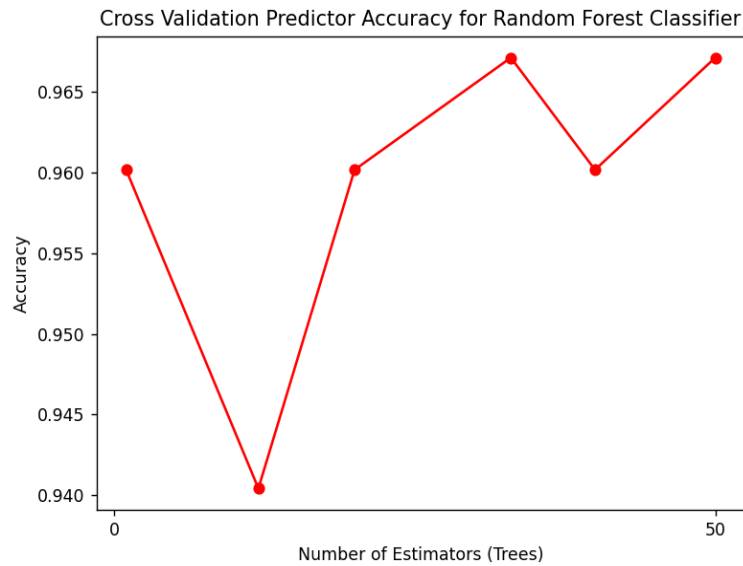
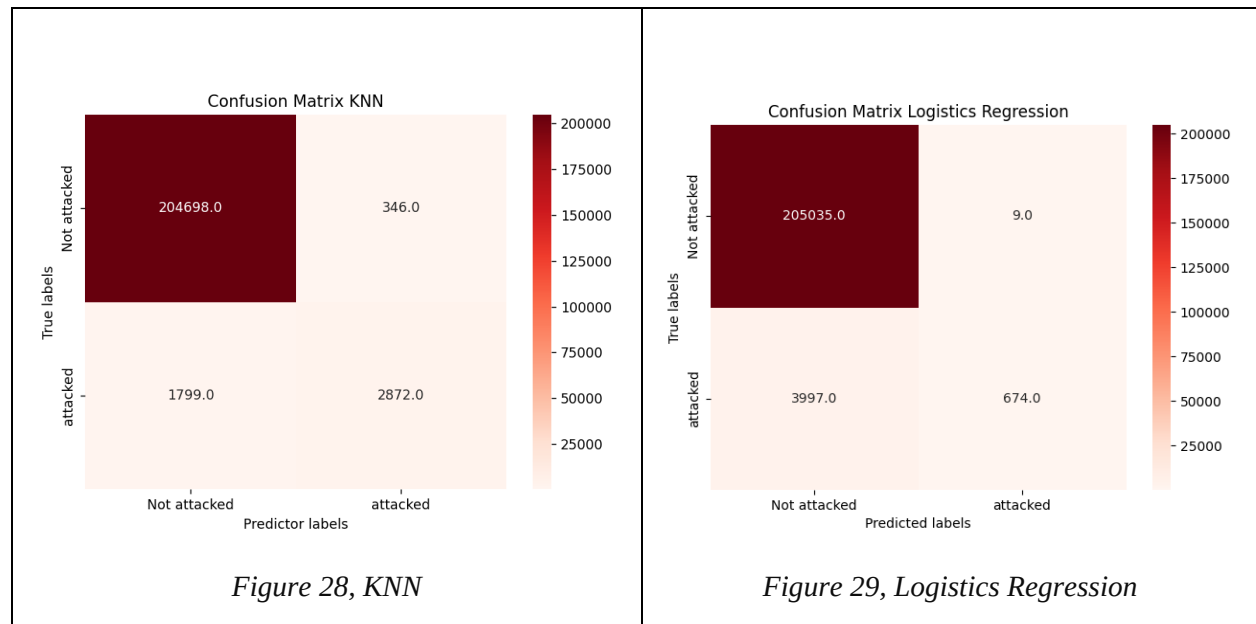


Figure 27, Random Forest

Confusion Matrix

The confusion matrix gives a summary as well as visualization of the results based on the number of true positive, true negative, false positive and false negative predictions. The table has 4 quadrants, and each quadrant is assigned a value representing the counts of each prediction type. It is used to define the performance of the classification model. It basically shows what kind of errors, if any. This can give pertinent information to determine if the model is confused or disorganized. Below are the results for KNN, Random Forest, and Logistic regression.



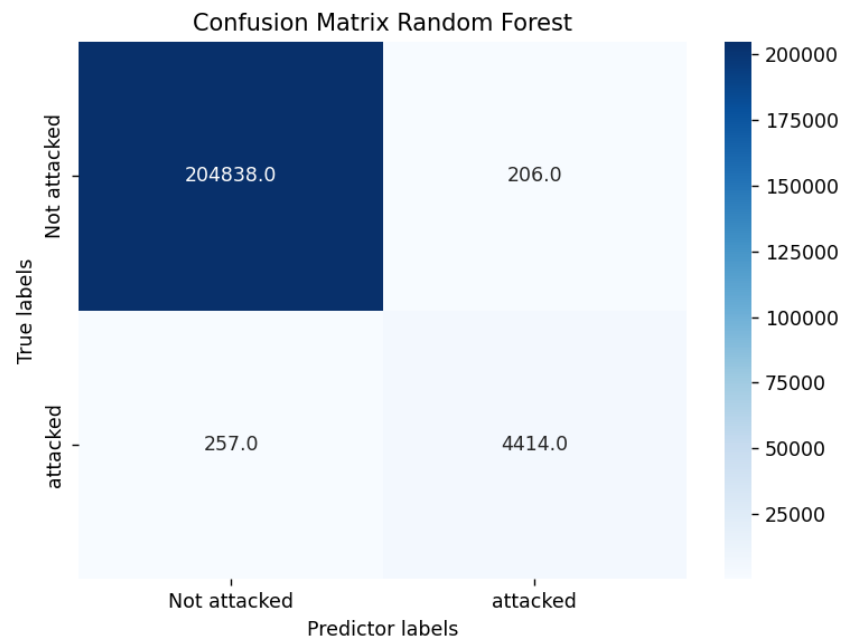


Figure 30, Random Forest

Normalized Confusion Matrix

The normalized confusion matrix enables visualization of how the labels are being predicted by the model. This is done to ensure class imbalance. This is different from the confusion matrix in that each row element is divided by the sum of the entire row.

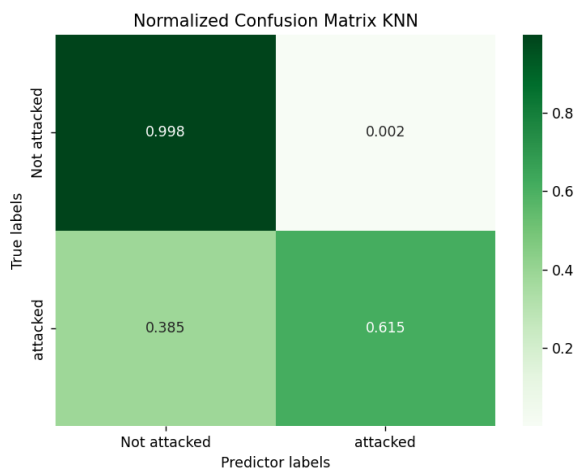


Figure 31, KNN

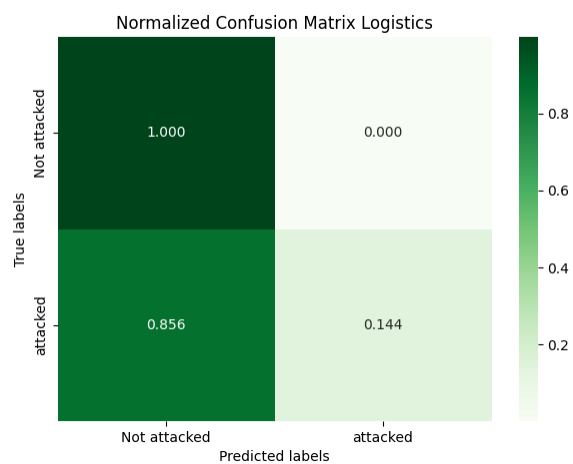


Figure 32, Logistics Regression

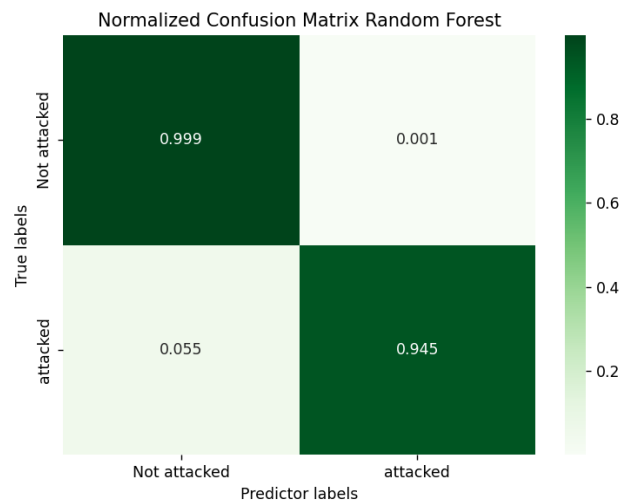


Figure 33, Random Forest

The RoC Curve

The Receiver Operating Characteristic Curve (RoC Curve) graphs the performance of the classification model. It has two parameters, true positive rates and false positive rates. The ROC plots the true positive rate and false positive rates at different entries. If the threshold is lowered more items will be positive. The Area Under the RoC Curve (AUC) is used to measure the entire two-dimensional areas underneath the entire RoC curve from points (0,0) to (1,1). AUC scores of 0.0 indicate that the predictions made by the model are 100% incorrect. AUC scores of 1.0 indicate that the predictions made by the model are 100% correct (Google, 2023)

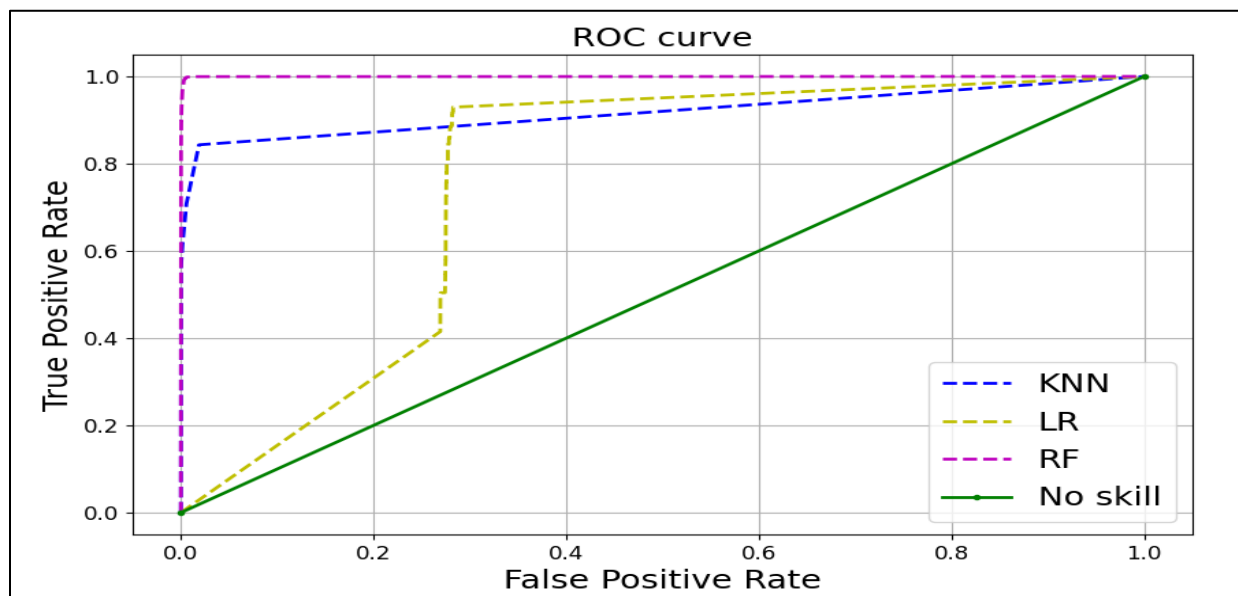


Figure 34, RoC curve

Precision Recall Curve

Precision and recall have a correlational relationship in that if you try to increase precision, recall will be affected negatively and the same is true in the opposite direction. Precision is a metric of how many correct positive predictions are truly correct. Recall is a measure of how many positive predictions were accurate. To evaluate the effectiveness of a model holistically, we must use both.

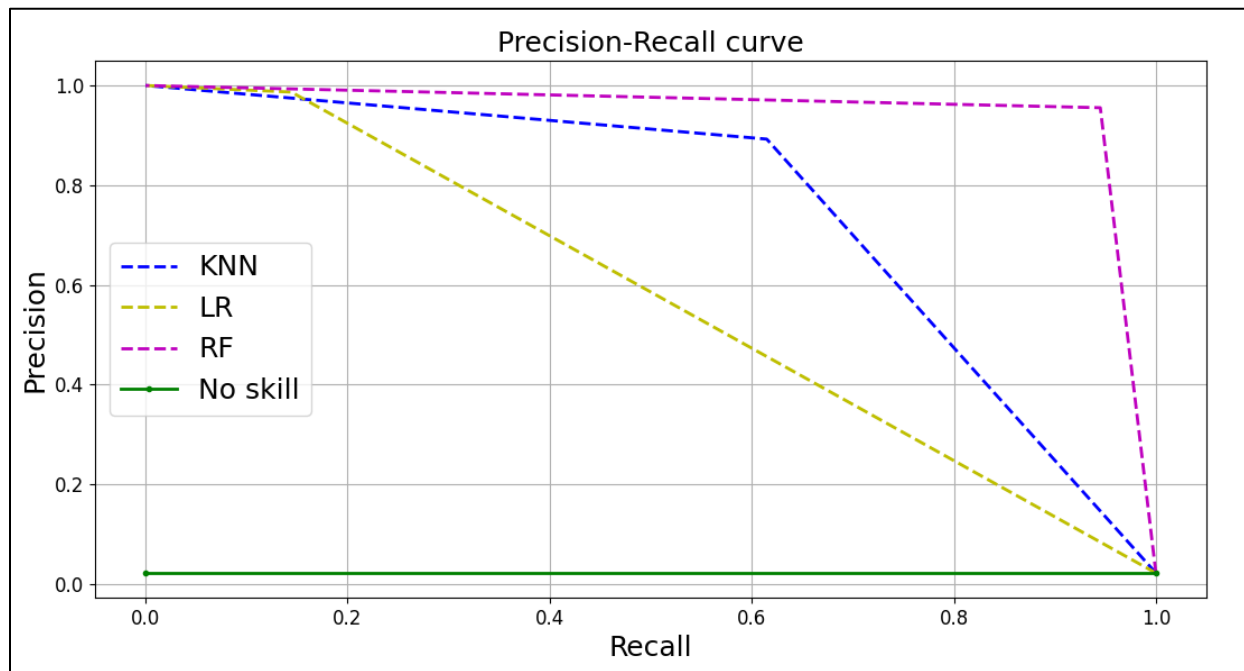


Figure 35, Precision recall curve

Your own comments:

Based on the results and all the evaluation method, we found that Random Forest classifier gives 99.78 accuracy, KNN gives 98.95% accuracy and Logistics regression gives 98.1% accuracy which is lowest. So, Random Forest gives more accuracy. Also, we evaluated the performance based on confusion and normalized confusion matrix and we found that false positive value in Random Forest is 0.055 which is very less than KNN (0.385) and Logistics Regression (0.856). So again, Random Forest classifiers performance is better than KNN and Logistics. Therefore, as per our observation, Random Forest gives better performance than other classifiers.

MODEL EVALUATED	KNN	RANDOM FOREST	LOGISTIC REGRESSION
PREDICTOR ACCURACY	98.95%	99.78%	98.1%
CROSS VALIDATION	97%	99%	98%
NORMALIZE CONFUSION MATRIX	FN VALUE 0.385	FN VALUE 0.055	FN VALUE 0.856

Table 4, Summarized results

References:

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4. Moustafa , N., & Slay, J. (2015). UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). *Military Communications and Information Systems Conference (MilCIS), Canberra, ACT*, 1-6. doi:10.1109/MilCIS.2015.7348942
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